

Mathematical Framework

Decision Support System for Sustainable Concrete Mix Optimization

1. Multi-Objective Optimization Problem

Concrete mix design requires simultaneous minimization of CO₂ emissions and material cost while maximizing compressive strength — three objectives that are inherently in conflict. This system applies a weighted multi-objective framework, enabling engineers to explore Pareto-optimal trade-offs interactively without requiring a single a priori preference statement.

2. Weighted Sum Objective Function

The three normalized objectives are aggregated into a single scalar score using the Weighted Sum Method. User-assigned weights (w_{CO_2} , w_{Cost} , $w_{Strength}$) reflect engineering priorities and must satisfy $w_{CO_2} + w_{Cost} + w_{Strength} = 1$, all $w_i \geq 0$. The composite objective to be minimized is:

$$F = w_{CO_2} \cdot f_{CO_2} + w_{Cost} \cdot f_{Cost} - w_{Strength} \cdot f_{Strength}$$

Equation 1: Weighted Sum Objective Function

The strength term carries a negative sign because maximizing compressive strength is equivalent to minimizing its negation within the minimization framework. All f_i terms are first normalized to [0, 1] before weighting (see Section 3).

Symbol	Objective	Direction	Unit
$w_{CO_2} \cdot f_{CO_2}$	CO ₂ Emissions	Minimize	kg/m ³
$w_{Cost} \cdot f_{Cost}$	Material Cost	Minimize	\$/m ³
$w_{Strength} \cdot f_{Strength}$	Compressive Strength	Maximize	MPa

Table 1: Objective Function Components

3. Min-Max Normalization

To ensure objectives on different measurement scales contribute proportionally to the composite score, each raw objective value x is normalized to the unit interval [0, 1] using Min-Max normalization before applying weights:

$$f_{norm}(x) = (x - x_{min}) / (x_{max} - x_{min})$$

Equation 2: Min-Max Normalization

Here x_{min} and x_{max} are the minimum and maximum values observed across all candidate mixes in the dataset. The result $f_{norm}(x) \in [0, 1]$ for any in-sample value, preventing any single objective from dominating due to magnitude differences.

4. Linear Interpolation for Mix Proportion Estimation

When a target performance point is selected on the Pareto front, the corresponding mix proportions (cement, slag, fly ash, water) are estimated by linear interpolation between the two nearest Pareto-optimal neighbors A and B in objective space:

$$\mathbf{x}_{\text{interp}} = \mathbf{x}_A + t \cdot (\mathbf{x}_B - \mathbf{x}_A), \quad t \in [0, 1]$$

Equation 3: Linear Interpolation for Mix Proportions

The parameter t is the normalized distance from A to the target point along the line segment AB. This provides a smooth, physically plausible estimate of feasible mix designs between discrete database entries without requiring a full surrogate model.

5. Engineering Constraints

The optimization operates subject to the following engineering and physical feasibility constraints:

- Minimum compressive strength: $f_{\text{Strength}} \geq f_{\text{min}}$ (user-defined, e.g. 25 MPa for structural RC)
- Water-to-cement ratio: $0.30 \leq w/c \leq 0.65$ (workability and durability limits per EN 206)
- Total binder content: $280 \leq (\text{Cement} + \text{Slag} + \text{Fly Ash}) \leq 550 \text{ kg/m}^3$
- Weight normalization: $w_{\text{C}} + w_{\text{S}} + w_{\text{FA}} + w_{\text{W}} = 1$, all $w_{\text{■}} \geq 0$
- Non-negative ingredient quantities: Cement, Slag, Fly Ash, Water $\geq 0 \text{ kg/m}^3$

DISCLAIMER — Decision Support Tool Only: This document describes the mathematical framework of a prototype system. All optimization results are based on simulated data and must be validated through physical laboratory testing and independent expert engineering review before application in any construction context.